High-resolution Coral Geodesy in the Solomon Islands

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Introduction & Objectives

The Problem: Modern geodetic data (GPS) is crucial for monitoring megathrust earthquakes but is temporally limited, with most records only beginning in the 1990s.

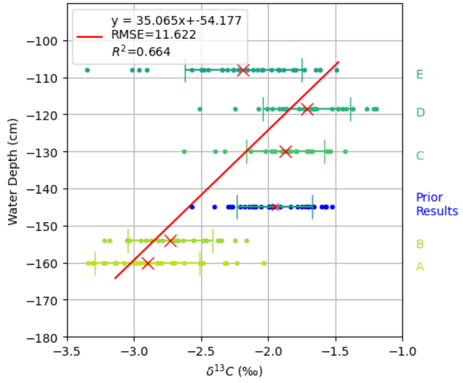
The Solution: We utilize coral skeletons as temporally high-resolved and purely vertical historical geodetic recorders in subduction settings.

Goal: To establish an annual-scale geodetic record that extends centuries back, providing a deeper understanding of the long-term earthquake cycle.

Geological Setting & Methodology

Solomon Islands

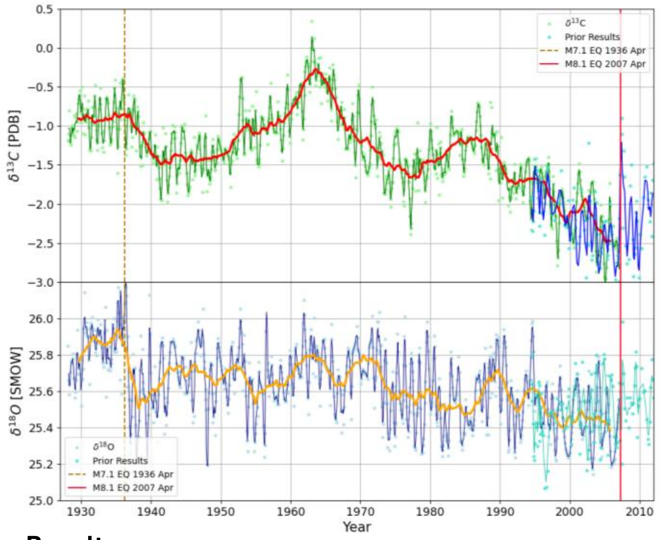
Study focused on the New Georgia Group, where the active trench is exceptionally close to the coastline (<30 km). This proximity makes the region highly sensitive to vertical deformation from subduction zone processes.



Methods

Sampling: Collected **575** horizontal and **141** vertical coral samples. We measured stable isotopes $\delta^{13}C$ and $\delta^{18}O$ to estimate relative sea level (RSL).

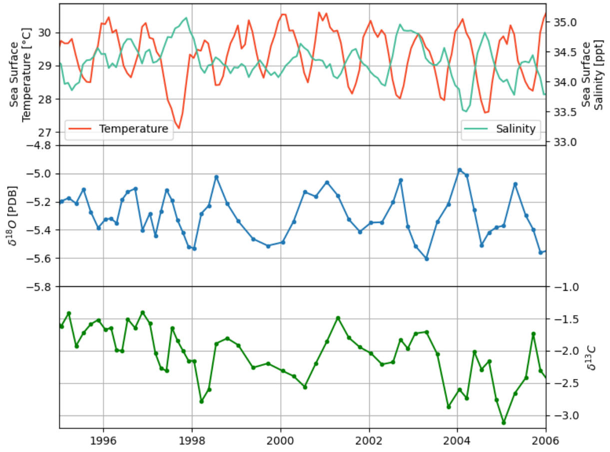
Calibration: Established a nearly linear relationship between water depth and $\delta^{13}C$ by analyzing vertical samples from synchronized ages.

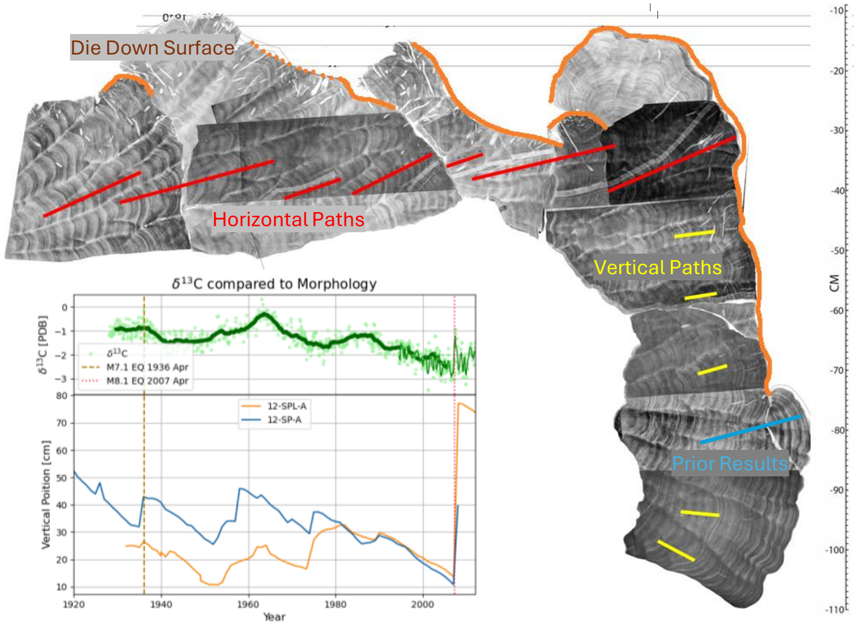


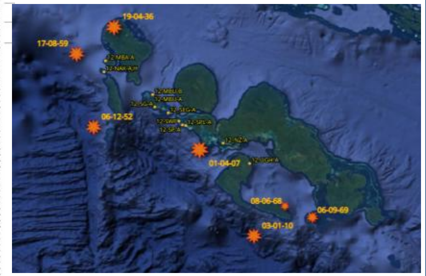
Results

Stable isotopes clearly record major regional seismic events:

- **1936 Event 7.0 M_W:** Captured primarily in the δ18O
- **2007 Event 8.1 M_W:** Strongly reflected in δ13C





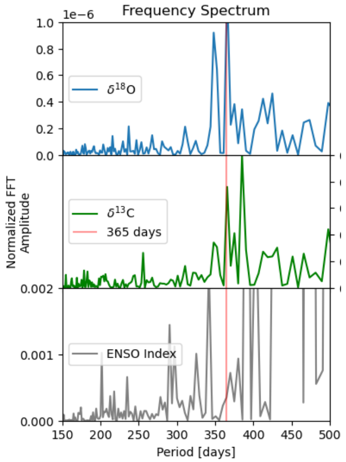


Decoupling Tectonic vs. Climatic Signals

- Isotopic signals are influenced by any local and global sea level change, Sea Surface Temperature (SST), Sea Surface Salinity (SSS), ENSO variability or Pacific Decadal Oscillation (PDO).
- By comparing coral data with external instrumental datasets, we successfully isolated the **tectonic signal** from climatic noise. Also spectral tools help to filter high frequency parameters compared to tectonics.

Morphology vs. Geochemistry

- While coral morphology indicates RSL change, it is often "binary". It only records events that reach the low-tide level. **Isotope analysis** provides a continuous vertical record, even when the coral remains submerged.



Summary & Conclusions

- δ13C correlates strongly with RSL and aligns with external environmental datasets (SST, SSS, and Tide Gauges).
- Coral morphology provides a structural "sanity check" that validates the isotopic trends.
- **Major Finding:** We demonstrate that δ13C is a robust indicator of vertical tectonics and serves as a powerful proxy for **paleogeodesy** in subduction environments.

Ongoing & Future Work

- Implementing a **bayesian age model** to refine the temporal resolution of the proxy.
- **Scaling:** Expanding the analysis to a broader suite of coral samples for a comprehensive regional reconstruction of the Solomon Islands' seismic history.